

<u>Variable</u>	<u>Type of variable</u>		<u>Model used to predict missing data*</u>	<u>Missing value</u> %
AGENDA ITEM				
<u>Mother's place of birth</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.1%</u>	
	<u>Binary</u>		<u>0.5%</u>	
<u>Health insurance coverage</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.5%</u>	
<u>Smoking before pregnancy</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.8%</u>	
	<u>Binary</u>		<u>0.6%</u>	
<u>Smoking during pregnancy</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.3%</u>	
	<u>Binary</u>		<u>3%</u>	
<u>Living with a partner</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>12.4%</u>	
<u>Parity</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>15.3%</u>	
<u>Number of ultrasounds</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>4.3%</u>	
	<u>Binary</u>		<u>2.1%</u>	
<u>Restrictive diet before pregnancy</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.6%</u>	
		<u>Logistic regression</u>		
<u>Restrictive diet during pregnancy</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>0.8%</u>	
<u>Gestational diabetes</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>2.7%</u>	
	<u>Binary</u>		<u>6.1%</u>	
<u>Hypertension</u>		<u>Logistic regression</u>		
<u>Psychological difficulties during pregnancy</u>	<u>Binary</u>	<u>Logistic regression</u>	<u>6.1%</u>	
		<u>Logistic regression</u>		
<u>Hospitalization during pregnancy</u>	<u>Binary</u>		<u>0%</u>	
	<u>(yes/no)</u>		<u>0.1%</u>	
<u>Premature rupture of membranes</u>	<u>Binary</u>	<u>Logistic regression</u>		
<u>Maternal medical history that can influence weight variations</u>		<u>Logistic regression</u>		
		<u>Logistic regression</u>		



Maternal medical
history that could
influence fetal
growth

Logistic
regression

Women with
child $\leq$ 2 years

Logistic
regression

Child's sex

Logistic
regression

Maternal age

Continuous

Linear
regression

0.02%

Maternal weight

Continuous

1.3%

Maternal height

Continuous

Linear
regression

0.03%

Body mass index

Continuous

Linear

1.3%

Gestational weight gain

Continuous

regression

2%

Gestational age at birth

Continuous

Linear
regression

0.8%

Birth weight

Continuous

1.4%

Birth length

Continuous

Linear
regression

9%

Head circumference at birth Ultrasonography
data second trimester (Z-scores)

Continuous

Linear

9.7%

Continuous

regression

25-69%**

Ultrasonography data third trimester (Z-scores)

Continuous

Linear
regression

20-69%**

Linear
regression

Linear
regression

Linear
regression

Linear
regression

Weight variations before pregnancy

Categorical (3
categories)

Multinomial
regression

13.2%

Activity status

Maternal level of education

2.4%



<u>Marital status</u>	<u>Categorical (3 categories)</u>	<u>Multinomial regression</u>	<u>0.01%</u>
<u>Socio-professional category</u>			<u>2%</u>
<u>Complementary health assurance</u>	<u>Categorical (4 categories)</u>	<u>Multinomial regression</u>	<u>0%</u>
<u>Medical history of diabetes</u>	<u>Categorical (3 categories)</u>	<u>Multinomial regression</u>	<u>2.1%</u>
<u>Medical history of hypertension</u>			<u>2.1%</u>
<u>Pregnancy caregiver</u>	<u>Categorical (5 categories)</u>	<u>Multinomial regression</u>	<u>1.4%</u>
<u>Prenatal consultations</u>			<u>0.9%</u>
<u>Consumption of alcohol</u>	<u>Categorical (3 categories)</u>	<u>Multinomial regression</u>	<u>1.8%</u>
<u>Maternity size strata</u>	<u>Categorical (4 categories)</u>	<u>Multinomial regression</u>	<u>0.7%</u>
<u>Period of inclusion</u>			<u>0%</u>
<u>Mode of birth</u>	<u>Categorical (3 categories)</u>	<u>Multinomial regression</u>	<u>0%</u>
	<u>Categorical (4 categories)</u>	<u>Multinomial regression</u>	<u>1.9%</u>
	<u>Ordinal (3 categories)</u>	<u>Multinomial regression</u>	
	<u>Categorical (4 categories)</u>	<u>Multinomial regression</u>	
	<u>Categorical (5 categories)</u>	<u>Multinomial regression</u>	
	<u>Categorical (4 categories)</u>	<u>Multinomial regression</u>	
	<u>Categorical (3 categories)</u>	<u>Multinomial regression</u>	

EMERGENCY AMBULANCE SERVICES COMMITTEE

EASC PERFORMANCE REPORT

Date of meeting

10/05/2022



FOI Status		Open/Public
If closed please indicate reason		Not Applicable—Public Report
Prepared by		Ross Whitehead, Deputy Chief Ambulance Services Commissioner
Presented by		Ross Whitehead, Deputy Chief Ambulance Services Commissioner
Approving Executive Sponsor		Chief Ambulance Services Commissioner
Report purpose		FOR DISCUSSION / REVIEW
Engagement (internal/external) undertaken to date (including receipt/consideration at Committee/group)		
Committee/Group/Individuals	Date	Outcome
ACRONYMS		
DAG	Delivery Assurance Group	
EMRTS	Emergency Medical Retrieval and Transfer Service	
NEPTS	Non-Emergency Patient Transport Service	
WAST	Welsh Ambulance Services NHS Trust	

1. —SITUATION/BACKGROUND

- 1.1—The purpose of the report is to provide an update on current emergency ambulance performance for Committee Members.
- 1.2—Further iterations of this report will include performance information for Non-Emergency Patient Transport Services and the Emergency Medical Retrieval and Transfer Service.



1.3—This report presents information in line with the most recent publications of the Ambulance Quality Indicators.

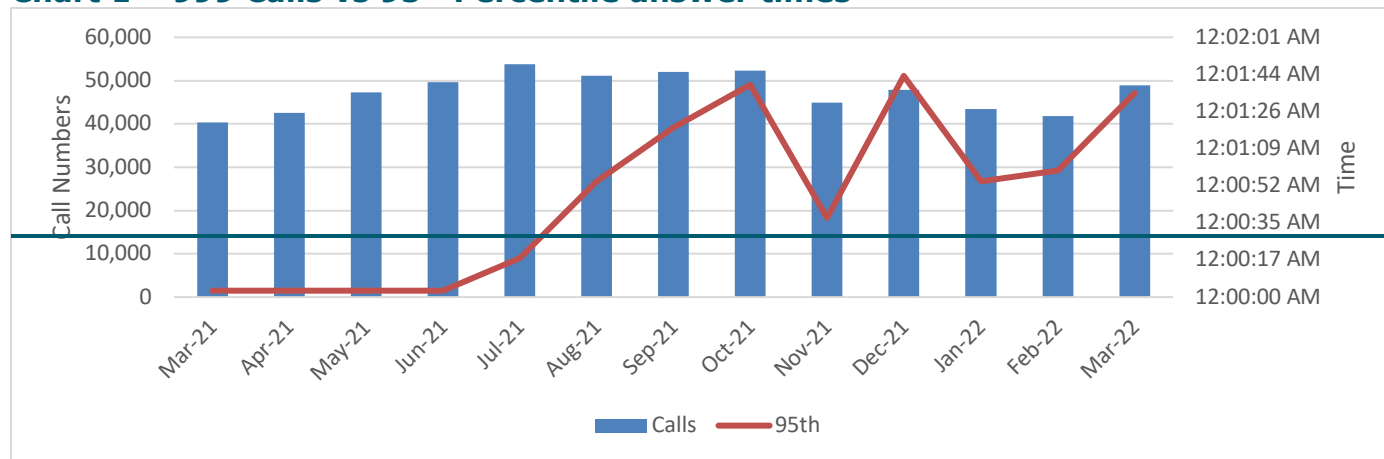
1.4—In response to feedback from Members, additional management information has been appended to this report to support discussions at the Committee meeting. This information is un-validated and subject to change prior to publication, therefore use of this information outside of the meeting should be treated with caution.

2. ~~SPECIFIC MATTERS FOR CONSIDERATION BY THIS MEETING (ASSESSMENT)~~

2.1—This report will set out activity and performance across Calls, Incidents, Response and Output.

2.2—The chart below shows call volume versus the 95th percentile answer time. Whilst the median and 65th percentile answer times continue to perform well at 2 and 3 seconds, there is a clear deterioration in the 95th percentile time.

Chart 1—999 Calls vs 95th Percentile answer times

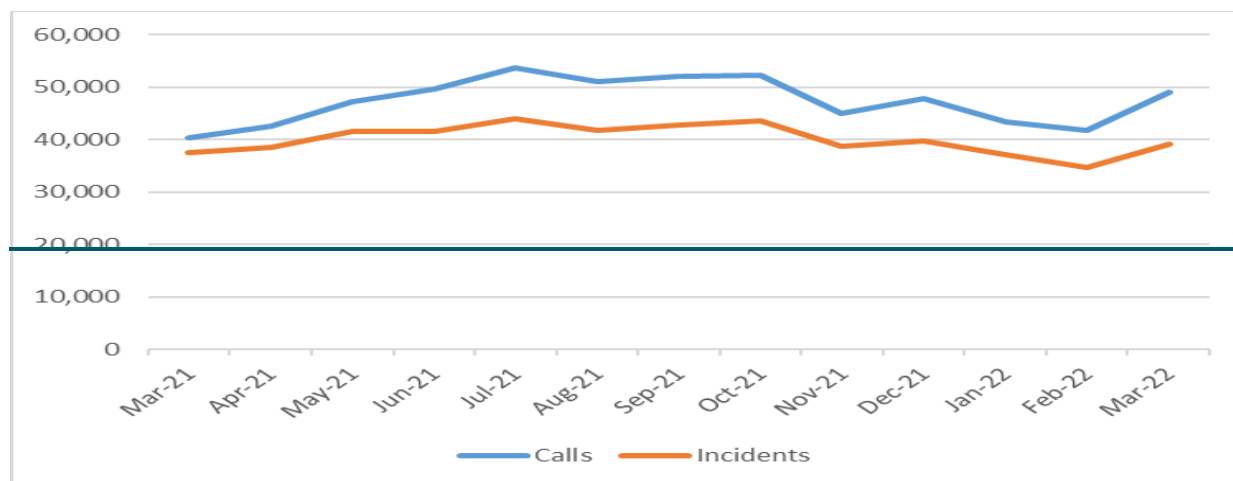


2.3—Not all calls received by the 999 clinical contact centres (CCCs) will generate an incident. The chart below shows the relationship between call volumes and incident generation.

2.4—There is a growing gap between the number of calls answered and the number of incidents generated, this is often seen at times of escalation related to individuals calling back due to a delayed response.



Chart 2—Calls v Incident volumes

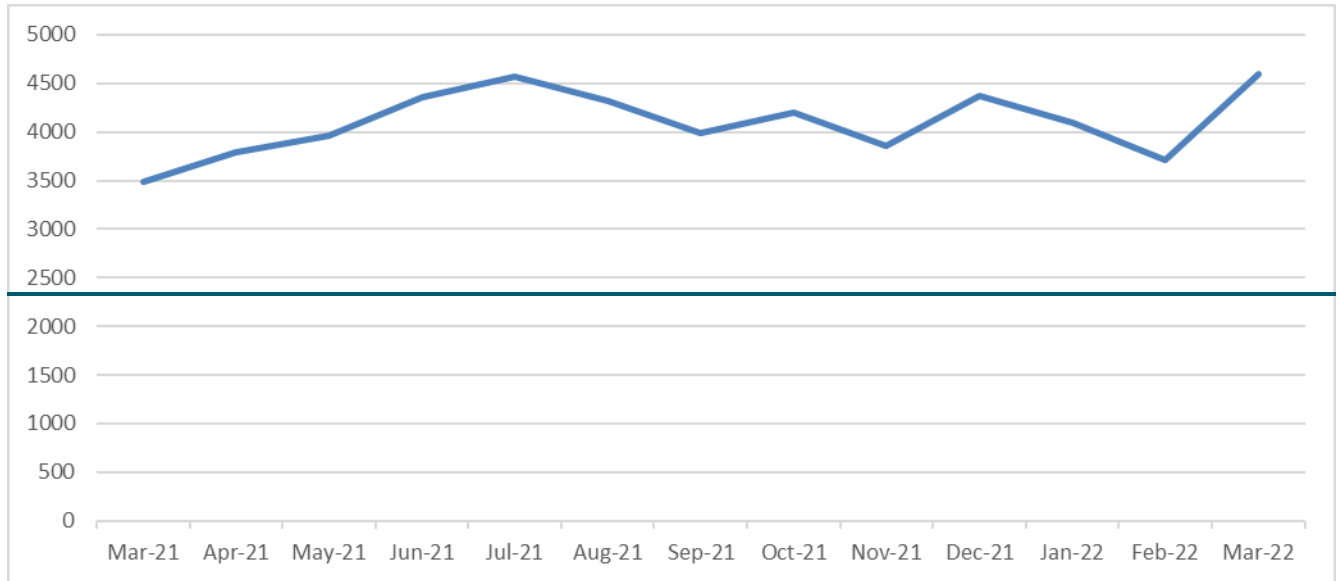


2.5—The Committee Members have recognised the importance of remote clinical support for patients and increasing the numbers of patients receiving an outcome of hear and treat.

2.6—The chart overleaf demonstrate the volume of incidents resolved by hear and treat. Recent investment in both staff and technology within the clinical support desk should support improvements in this level as well as more granular data on the outcomes for patients.

2.7—This chart contains information for 999 calls resolved by the emergency ambulance services clinical support desk as well as incidents passed to 111 for resolution.

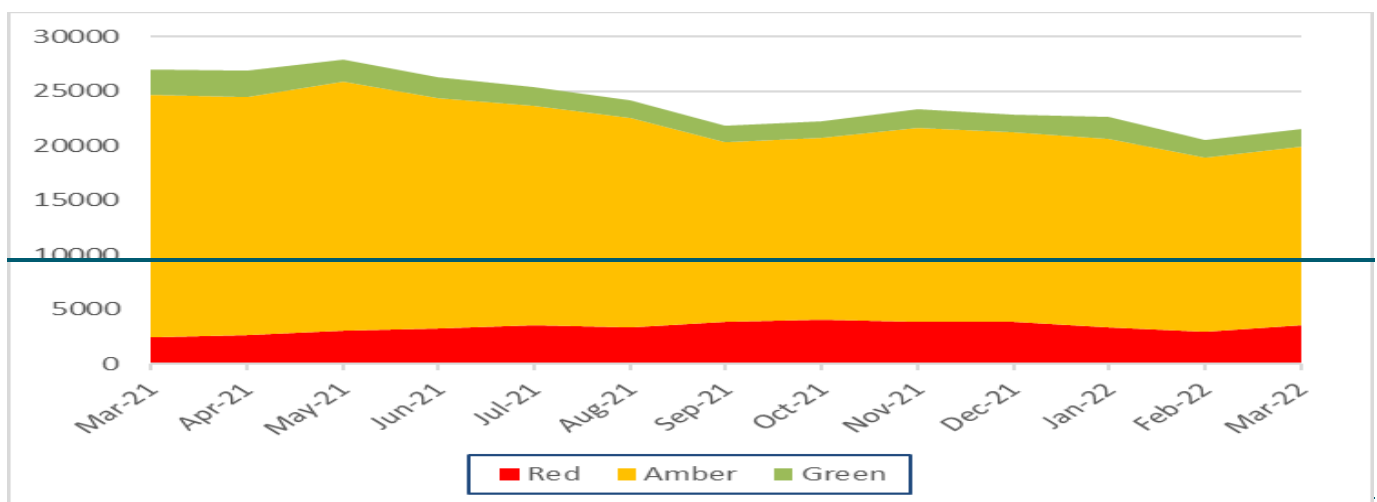
Chart 3—Hear and Treat Volumes.



2.8 The chart below demonstrates the categories of incidents receiving at least 1 ambulance response. Whilst there is a clear reduction in the number of incidents receiving a response overall, it should be noted that much of this reduction is driven by escalation actions.

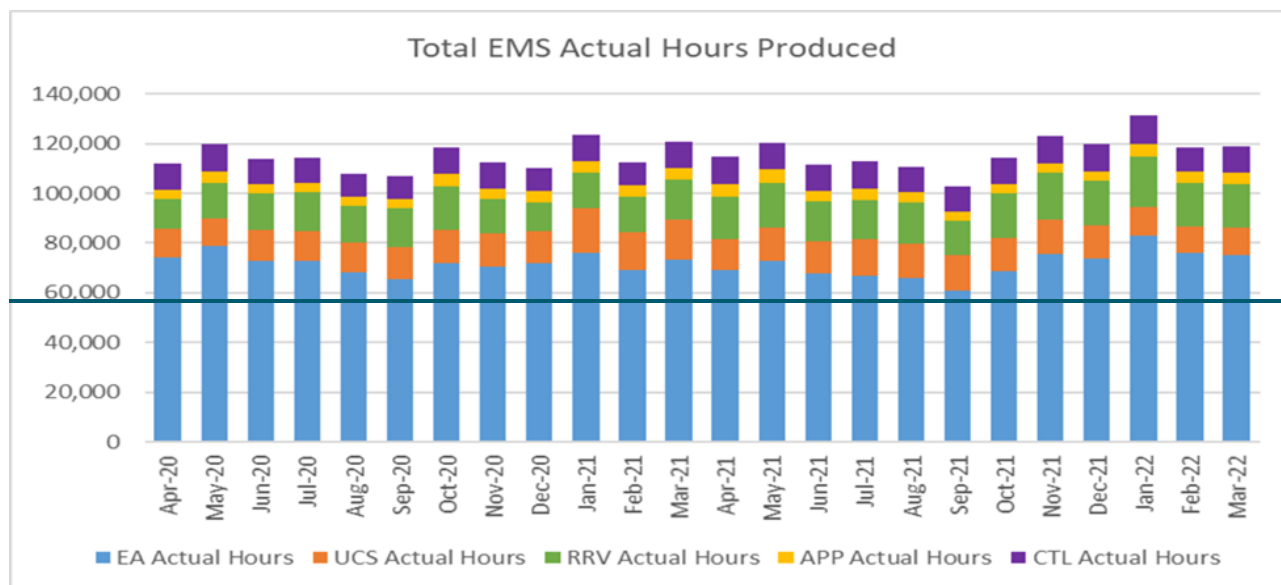
2.9 Committee Members should also note the increase in Red incident volume. By their nature, red incidents often require multiple responses at scene and so have a disproportionate effect on resource availability for other incidents.

Chart 4 Categories of incidents receiving a response





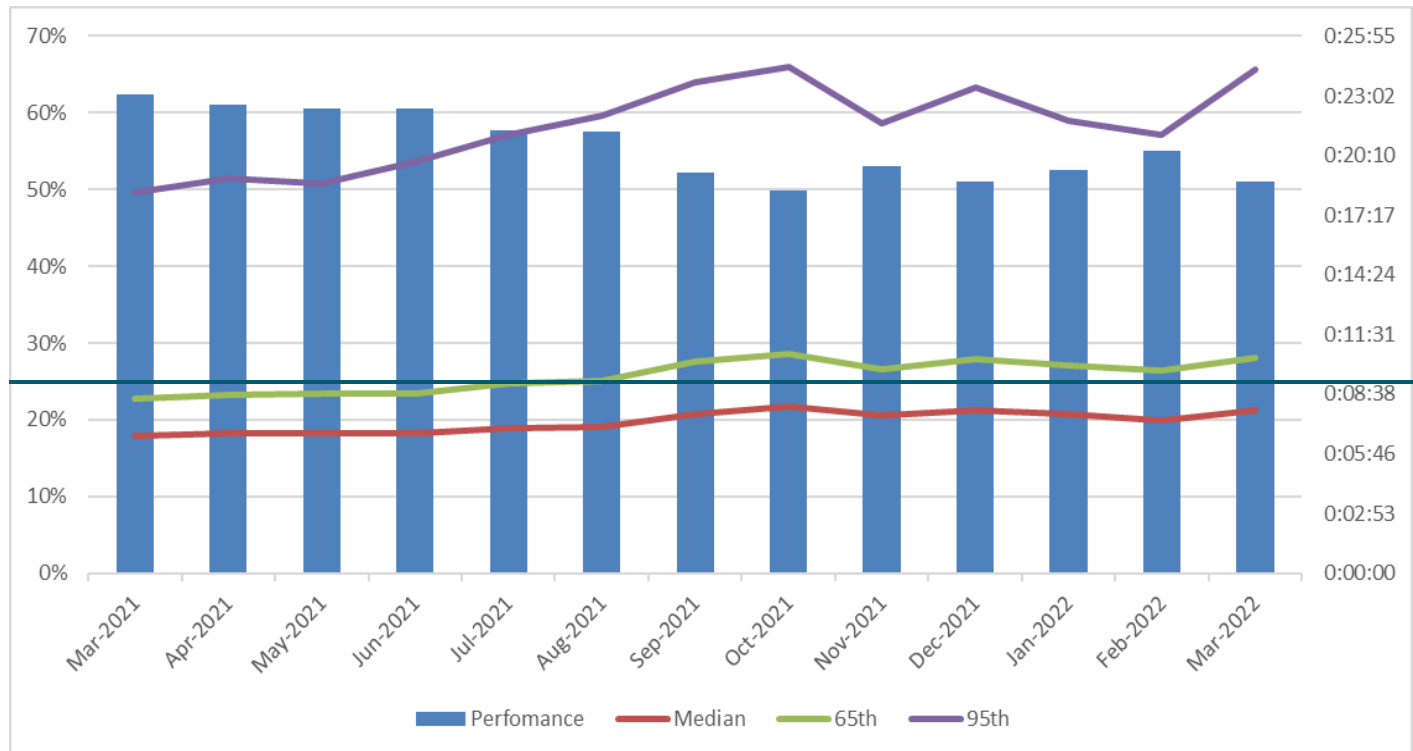
2.10 At the last meeting, Members requested information on the number of hours produced by the ambulance service each month. The chart below demonstrates this.



2.11 The chart overleaf demonstrates Red performance alongside the range of response times that have been achieved. The lack of delivery of the target has been well documented, however committee members should note the increasing response times that underpin this.



Chart 5 – Red Response and Performance

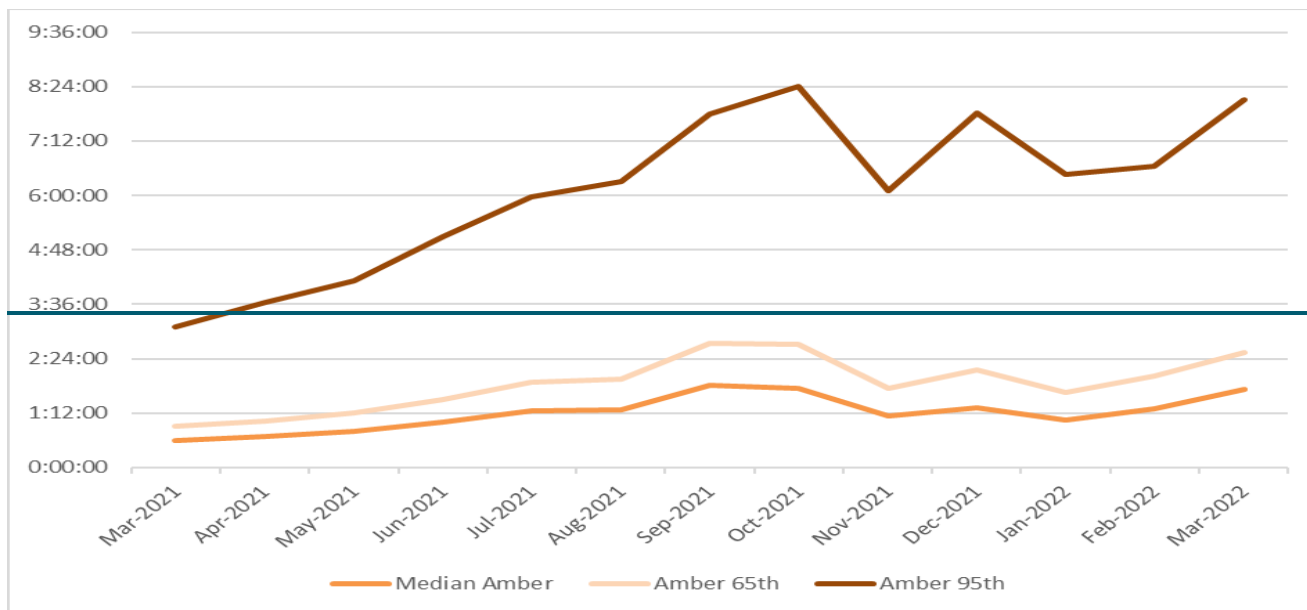


2.12 The chart below shows the response times for Amber patients. Whilst there is no time target for Amber patients, it's accepted that these patients still require a timely response. The current response times for this category are significantly above the level that would be considered appropriate.

2.13 The key to delivery of appropriate response within Amber is the availability of ambulance resource. The development of the ambulance utilisation metric will be a key enabler for the committee and operational teams to understand and improve this area of response.

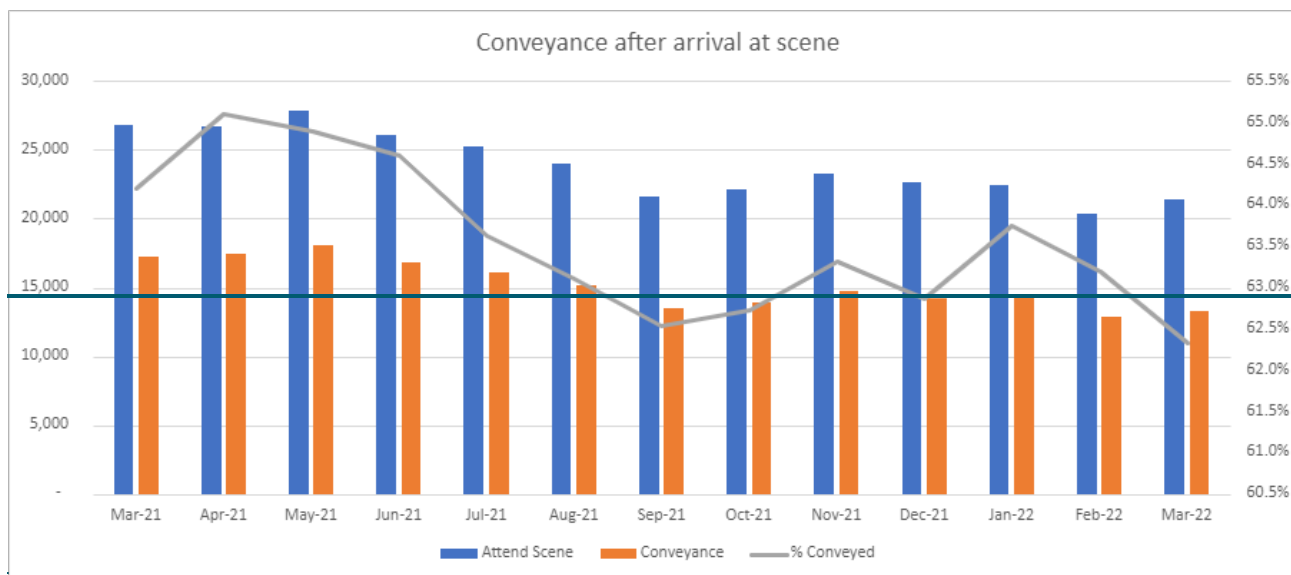


Chart 6—Amber responsiveness



2.14 Optimising conveyance is a key aim of the urgent and emergency care programme. Whilst the chart below shows a reduction in both conveyance volume and percentage, this impact must be considered in light of the information provided earlier in this report which showed the reduction in attendance in response to escalation decisions.

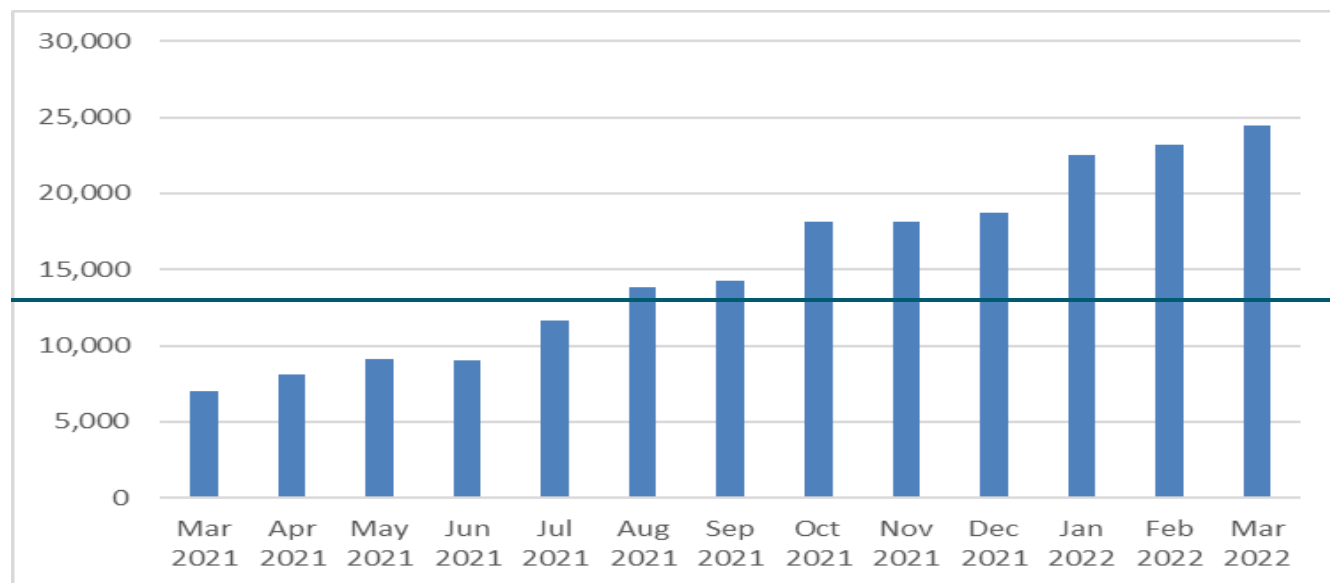
Chart 7—Conveyance volumes





2.15 The chart below demonstrates the level of lost hours to handover outside emergency departments (ED) in Wales.

Chart 8 — Handover Lost Hours



3. — KEY RISKS/MATTERS FOR ESCALATION TO THE COMMITTEE

3.1 Committee Members will note from the information within this report that there has been a deterioration in a range of performance indicators.

3.2 Due to the time sensitive nature of some type of incidents presenting to the ambulance service, response performance can be used as a proxy for harm and as such the committee should anticipate increase levels of harm and reductions to patient safety if these performance trajectories are not reversed.

4. — IMPACT ASSESSMENT

Quality/Safety/Patient Experience implications	There are no specific quality and safety implications related to the activity outlined in this report.
Related Health and Care standard(s)	Governance, Leadership and Accountability



Equality impact assessment completed	Not required
Legal implications / impact	There are no specific legal implications related to the activity outlined in this report.
Resource (Capital/Revenue £/Workforce) implications / Impact	There is no direct impact on resources as a result of the activity outlined in this report.
Link to Commissioning Intentions	The Committee's overarching role is to ensure its Commissioning Strategy for Emergency Ambulance Services utilising the five step patient pathway outlined within the National Collaborative Commissioning Quality and Delivery Agreement and the related outcomes for each care standard aligned with the Institute of Healthcare Improvement's (IHI) 'Quadruple Aim' are being progressed.
Link to Main WBFG Act Objective	Commitment to corporate social responsibility and improving health & social equity, work with our staff, partners and communities to build strong local relationships and solid foundations of the past

5. RECOMMENDATION

5.1 The Emergency Ambulance Services Committee is asked to:

- **NOTE** the content of the report.
- **DISCUSS** additional actions that the committee could take to improve performance delivery of commissioned services

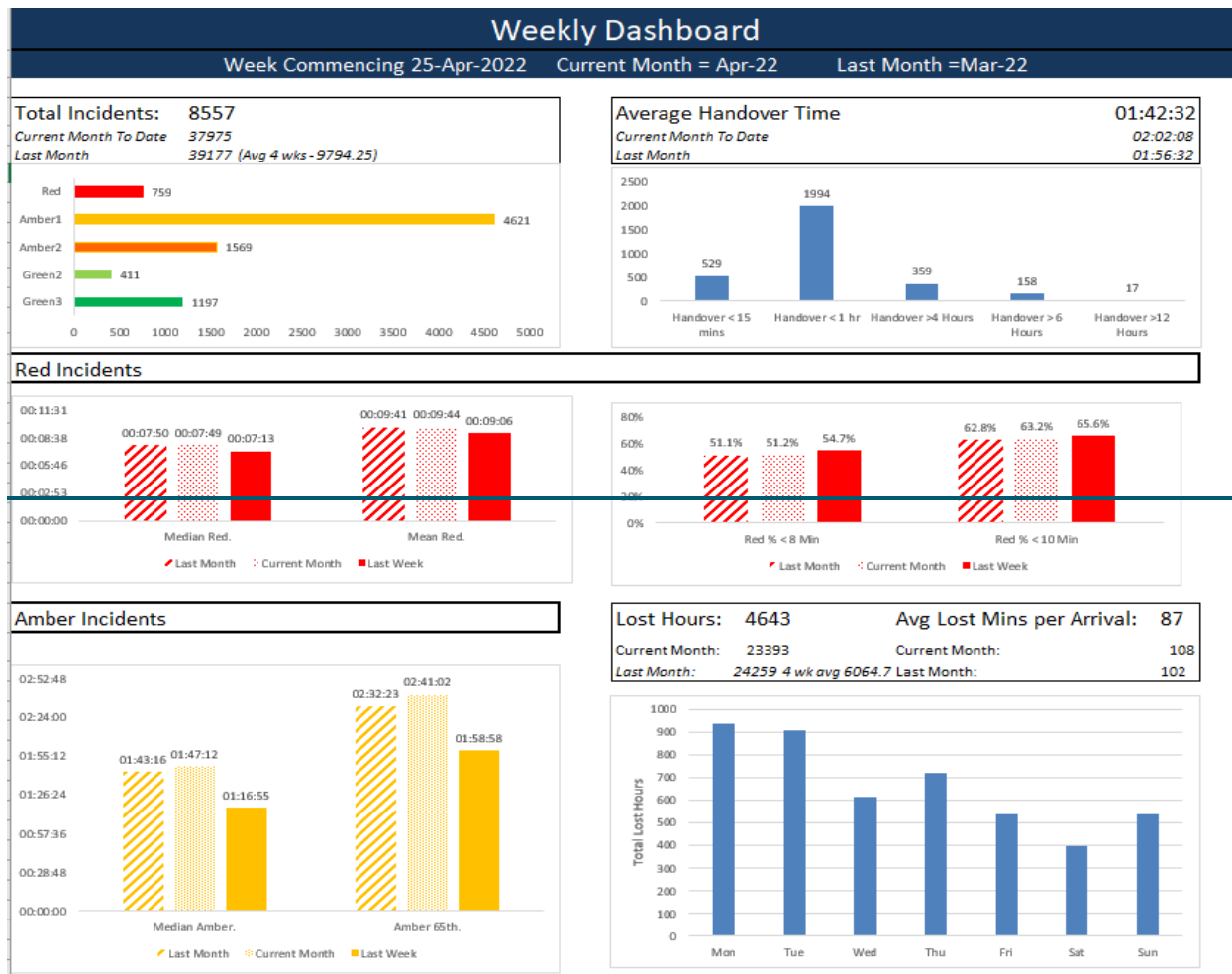


GIG
CYMRU
NHS
WALES

Pwyllgor Gwasanaethau
Ambiwylans Brys
Emergency Ambulance
Services Committee



**MANAGEMENT INFORMATION FOR DISCUSSION – THIS INFORMATION SHOULD NOT BE USED
OUTSIDE OF THE MEETING AND IS SUBJECT TO CHANGE**



Key Observations - NHS Wales:

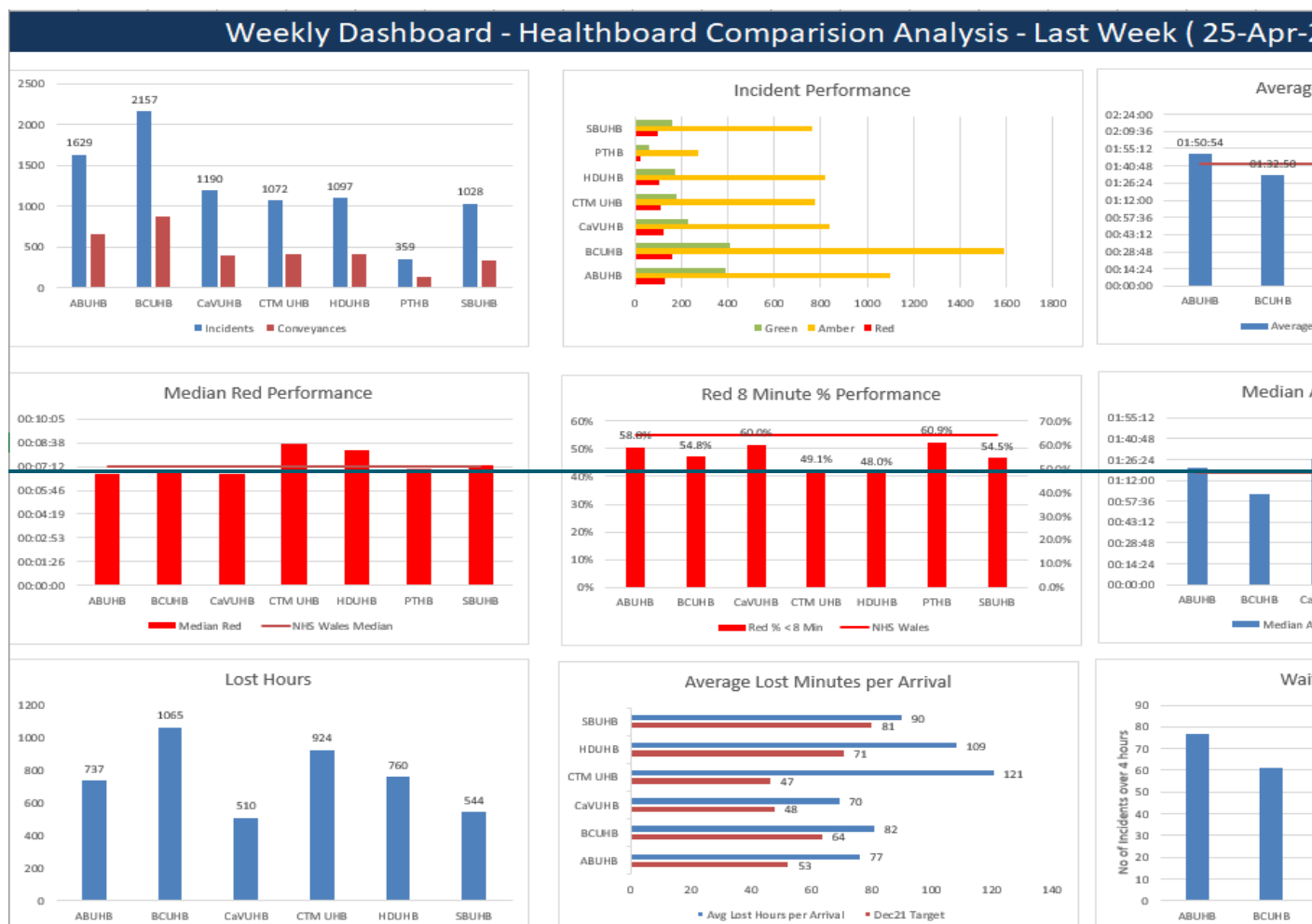
Avg Handover	Avg Handover for last week has improved by 27% (01:42:00) as compared to previous week (02:20:00)
Red Incidents	Median Red has improved by 8% as compared with the previous week. Red 8 min % slightly improved last week (54.7%), as compared to last week (51.4%)
Amber Incidents	Median Amber was 33% better last week as compared to previous week
Lost Hours	Lost Hours is 1449 (24%) lower for the last week as compared to previous week
Avg Lost Minutes	Last week is lower (86 mins) as compared to previous week (128 mins)



GIG
CYMRU
NHS
WALES

Pwyllgor Gwasanaethau
Ambiwlans Brys
Emergency Ambulance
Services Committee

**MANAGEMENT INFORMATION FOR DISCUSSION—THIS INFORMATION SHOULD NOT BE USED
OUTSIDE OF THE MEETING AND IS SUBJECT TO CHANGE**



**MANAGEMENT INFORMATION FOR DISCUSSION—THIS INFORMATION SHOULD NOT BE USED
OUTSIDE OF THE MEETING AND IS SUBJECT TO CHANGE**

Key Observations	
Handover Rates	All HBs, except AB improved their handover rates last week, as compared to last month. BCU saw the biggest improvement (26%) last week, as compared to last month.
Red 8 min %	All HBs, except C&V, saw an improvement in their red 8 min %. C&V saw a deterioration. Powys and BCU saw the biggest improvements last week (25% and 21% respectively).
Avg Lost Mins per Arrival	All HB's except AB saw an improvement in Avg lost mins last week, compared to last month. SB and BCU saw improvements of 22% and 24% respectively. AB had a deterioration of 1%.
Lost Hours	All HBs saw a reduction in their lost hours last week, compared to previous week. BCU had the biggest reduction (34%), whilst AB had the smallest reduction (1%).